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Rethinking Standards of Living Beyond GDP: A Dynamic Panel Approach to Poverty, Inequality, and Development

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The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam.

Abstract

This study analyzes the clustering of 15 countries based on GDP per capita and poverty gap, using models with and without inequality adjustment. Time-varying coefficients are estimated via Jia Chen’s approach, and missing data are addressed through cubic interpolation. Results show developing countries cluster together in GDP models, while high-income countries group distinctly. In poverty gap models adjusted for inequality, high-income countries cluster closely, whereas developing countries display diverse patterns. Findings highlight the varying impact of inequality on economic and poverty outcomes, emphasizing the need for tailored policies. Limitations due to sample size are acknowledged, with suggestions for future research.

1 Introduction

At some point, we’ve all paused to ask: *What does growth really mean for a society?* Is it just about numbers on a ledger—GDP rising year after year? Or is there something deeper, something that captures whether people’s lives are truly improving? This question has fascinated economists and social scientists for centuries. Since the dawn of recorded history, humans have sought that “Eureka!” moment—like Newton’s gravity insight—that would unlock the secret formula for progress (Stiglitz et al., 2010).

In ancient times, with few ways to measure well-being, societies relied on simple output measures. If more goods and services were produced, it was assumed the economy was doing well (Maddison, 2001). Over time, economists developed models to explain how investments, savings, and inflation affected national output (Keynes, 1936). These ideas gave rise to Keynesian economics, which taught us that government spending can stabilize demand swings and influence employment and inflation (Blinder & Solow, 1973). Yet the tidy relationship between inflation and unemployment, famously captured by the Phillips Curve, unraveled in the 1970s, as economies faced stagflation (Friedman, 1977; Phelps, 1967).

This opened the door for Monetarism, which emphasized controlling the money supply to achieve stability (Jr., 1976). But even this theory faced real-world limitations, as monetary targeting often failed to predict bank behavior (Taylor, 1993). Then came the Solow Growth Model, which viewed growth as the result of capital, labor, and technological progress (Solow, 1956). While elegant, its convergence predictions often fell short in explaining persistent inequality between countries (Easterly, 2001).

More recently, development economists have challenged the assumption that GDP alone captures societal progress. Rising GDP may mask inequality, exclusion, and poor public services (Stiglitz et al., 2010; Alkire & Foster, 2011). Development, they argue, is not merely about raising output but about improving people’s lives—especially those most at risk of being left behind. Measures like the Human Development Index (HDI), the poverty gap, and the Gini coefficient help us understand who benefits from growth, and who doesn’t (United Nations Development Programme (UNDP), 2020).

Indeed, it is common to see countries with impressive GDP figures still struggling with deep poverty. The poverty gap reveals how far the poor fall below a basic standard of living, while the Gini coefficient shows how unevenly income is distributed (Deininger & Squire, 1996). These

outcomes are shaped not just by markets, but by governance—how governments spend public funds, and how much corruption diverts those funds away from productive use (Mauro, 1995; Kaufmann et al., 2010). Government budgets reflect a society’s values, and their allocation to health, education, or infrastructure can determine whether growth translates into broad-based well-being (Tanzi & Davoodi, 1997). Yet, when corruption enters the equation, the effectiveness of public spending is undermined (Transparency International, 2020).

Despite the complexity of these dynamics, many existing studies treat countries as homogeneous, assuming they respond uniformly to development drivers. But historical, institutional, and structural contexts matter. What works in one country may fail in another. This thesis takes a different approach.

Rather than asking only how GDP grows in response to factors like government spending, corruption, or HDI, we flip the lens to ask: *How does poverty itself react to these same drivers?* On the development side, we explore how human development, public expenditure, and governance quality affect poverty levels and inequality. On the structural side, we examine how these same factors influence GDP growth. This allows us to assess whether countries that respond similarly in economic terms also exhibit similar responses in poverty reduction.

To take this a step further, we ask: *Do these effects remain consistent when we consider inequality explicitly?* By introducing interaction terms with the Gini coefficient, we examine whether inequality conditions or reshapes the impact of key drivers on poverty. In other words, does the same level of government spending reduce poverty equally in a low-inequality society and a high-inequality one? Or does inequality fundamentally alter the development trajectory?

Using a panel dataset of countries over time, we apply time-varying coefficient models with kernel smoothing and incorporate latent group structures. This setup allows us to detect not only how the effects of these variables change over time but also whether clusters of countries respond in fundamentally different ways. This is especially relevant in a post-pandemic world, where economic shocks and governance challenges have widened development disparities.

Unlike traditional fixed-effects models, our framework captures evolving, non-linear relationships and heterogeneity across nations. Kernel smoothing highlights persistent trends over time, while latent group modeling uncovers hidden clusters of countries that share similar development dynamics—without forcing arbitrary groupings. This enables a more precise understanding of which policies work, when, and where.

In sum, this thesis aims to provide a deeper, more nuanced understanding of how poverty and inequality evolve in response to structural and policy variables—and how these relationships shift in the presence of inequality. It asks whether economic development truly reaches those who need it most, or whether inequality and governance stand in the way. By moving beyond static models and homogeneous assumptions, this research contributes to a more inclusive and actionable theory of development.

2 Methodology

Regression models are fundamental tools in economic research, especially for analyzing macroeconomic time series data. Traditionally, these models focus on a single entity, such as a country,

over time to understand how explanatory variables influence economic outcomes (Stock & Watson, 2019). However, this approach overlooks important interdependencies across countries or regions, which have become increasingly relevant in today’s globalized economy.

One straightforward approach is to run separate regressions for each country. Although simple, this method ignores cross-sectional dependencies like common shocks or spillover effects, which frequently occur in the macroeconomic landscape (Baltagi, 2021). Because economic agents operate within interconnected systems, their behaviors and outcomes tend to be interrelated. Univariate time series models cannot fully capture these cross-country interactions.

Panel data econometrics offers a more appropriate framework by combining the cross-sectional and time-series dimensions. This allows the model to account for differences between countries as well as dynamics over time (Hsiao, 2014). The simplest panel data models assume constant coefficients across all countries and time periods (Wooldridge, 2010). While computationally convenient, this assumption is often unrealistic, as the effects of explanatory variables may vary substantially across countries or evolve over time.

To address this limitation, researchers have developed heterogeneous panel models, which allow coefficients to vary either across units, over time, or both (Pesaran & Smith, 1995). Nevertheless, structural breaks—such as the 2008 global financial crisis or the COVID-19 pandemic—pose challenges for these models by causing shifts in the relationships between variables (Stock & Watson, 2002). In these situations, the magnitude and even the direction of effects may change as a response to wider economic disruptions.

Time-varying coefficient models provide a flexible approach to capture these dynamic relationships. One popular method is sieve estimation, which uses basis expansions like splines to approximate time-varying coefficients (X. Chen & Christensen, 2018). Despite its theoretical appeal, sieve methods can be computationally intensive and require careful tuning of basis functions and their derivatives.

An alternative, proposed by Chen (J. Chen, 2020), employs kernel smoothing techniques for nonparametric estimation of time-varying coefficients. Kernel smoothing is a well-established technique in nonparametric regression that allows smooth estimation of functional relationships without imposing restrictive parametric assumptions (Fan & Gijbels, 1996). The Epanechnikov kernel is particularly suitable because of its optimal bias-variance trade-off and good boundary properties (Epanechnikov, 1969).

Under standard regularity conditions, including mixing and moment assumptions, the kernel-based estimator is consistent and enjoys desirable asymptotic properties (J. Chen, 2020). This thesis adopts Chen’s kernel-based time-varying coefficient model to investigate how the effects of key variables on poverty evolve across countries and over time.

The remainder of this section is organized as follows. Section 2.1 introduces the formal specification of the time-varying coefficient model. Section 2.2 describes the kernel smoothing estimation procedure. Section 2.3 presents the latent group identification approach and the selection criteria for the number of groups and bandwidth parameter.

2.1 Model Specification

Consider panel data $\{(Y_{it}, X_{it}) : i = 1, \dots, N; t = 1, \dots, T\}$, where Y_{it} is the dependent variable and X_{it} is a p -dimensional vector of regressors. We model the data using a heterogeneous time-varying coefficient framework:

$$Y_{it} = \alpha_i + X_{it}^\top \beta_{it} + \epsilon_{it}, \quad (1)$$

where α_i captures entity-specific fixed effects, β_{it} is a vector of coefficients varying across entities and over time, and ϵ_{it} is an error term which may exhibit cross-sectional dependence but is assumed stationary over time.

Following the existing literature (Robinson, 1989; Cai et al., 2007), we assume that β_{it} can be represented as a smooth function of scaled time:

$$\beta_{it} = \beta_i \left(\frac{t}{T} \right), \quad (2)$$

where $\beta_i(\cdot)$ is a smooth, p -dimensional function unique to each entity i .

To incorporate a time-varying intercept, we redefine the regressors as

$$X_{it} = (1, X_{it,1}, \dots, X_{it,p-1})^\top,$$

and correspondingly write

$$\beta_i \left(\frac{t}{T} \right) = \left(\beta_{i,0} \left(\frac{t}{T} \right), \dots, \beta_{i,p-1} \left(\frac{t}{T} \right) \right)^\top.$$

Substituting into equation (1), the model becomes

$$Y_{it} = \alpha_i + \beta_{i,0} \left(\frac{t}{T} \right) + \sum_{k=1}^{p-1} \beta_{i,k} \left(\frac{t}{T} \right) X_{it,k} + \epsilon_{it}. \quad (3)$$

Because both α_i and $\beta_{i,0}(t/T)$ contribute to the intercept term, we impose the identifying restriction

$$\sum_{t=1}^T \beta_{i,0} \left(\frac{t}{T} \right) = 0,$$

which ensures the time-varying intercepts are mean zero and separate from the fixed effects (Boneva, 2015).

2.2 Estimation Procedure

The time-varying coefficient functions $\beta_i(\cdot)$ are estimated via kernel smoothing (J. Chen, 2020). For a given normalized time point $u \in (0, 1)$, the kernel-smoothed estimator for entity i is given by

$$\hat{\beta}_i(u) = \left[\sum_{t=1}^T X_{it} X_{it}^\top K \left(\frac{t - uT}{Th} \right) \right]^{-1} \left[\sum_{t=1}^T X_{it} Y_{it} K \left(\frac{t - uT}{Th} \right) \right], \quad (4)$$

where $K(\cdot)$ is the Epanechnikov kernel function, and $h > 0$ is the bandwidth controlling the smoothness of the estimated functions. This estimator provides a smooth trajectory $\hat{\beta}_i(u)$ capturing the evolution of coefficients over time, with weights determined by the proximity of each time point t to uT .

2.3 Latent Group Identification and Group-Level Estimation

While allowing for fully individual-specific time-varying coefficients is flexible, it may be inefficient when the number of entities N is large. To improve estimation efficiency and interpretability, we identify latent groups of entities with similar coefficient dynamics.

To do so, we compute pairwise distances between entities based on their estimated coefficient curves:

$$\hat{\delta}_{i,j} = \frac{1}{T} \sum_{t=1}^T \left\| \hat{\beta}_i \left(\frac{t}{T} \right) - \hat{\beta}_j \left(\frac{t}{T} \right) \right\|_q W \left(\frac{t}{T} \right), \quad (5)$$

where $\|\cdot\|_q$ denotes a vector norm (typically L_2) and $W(\cdot)$ is a weight function that may emphasize specific time periods.

Using the distance matrix $\hat{\delta}$, clustering algorithms (such as hierarchical clustering or k -means) are applied to partition entities into K latent groups. The optimal number of groups is selected based on information criteria.

After grouping, we estimate group-level time-varying coefficients by pooling data from all entities within each group:

$$\hat{\gamma}_{k|K}(u) = \left[\sum_{i \in \hat{G}_{k|K}} \sum_{t=1}^T X_{it} X_{it}^\top K \left(\frac{t - uT}{Th} \right) \right]^{-1} \left[\sum_{i \in \hat{G}_{k|K}} \sum_{t=1}^T X_{it} Y_{it}^c K \left(\frac{t - uT}{Th} \right) \right], \quad (6)$$

where Y_{it}^c denotes the dependent variable with individual fixed effects removed, and $\hat{G}_{k|K}$ is the set of entities assigned to group k .

The final grouped panel model is

$$Y_{it} = \alpha_i + \gamma_{k|K,0} \left(\frac{t}{T} \right) + \sum_{j=1}^{p-1} \gamma_{k|K,j} \left(\frac{t}{T} \right) X_{it,j} + \epsilon_{it}, \quad i \in \hat{G}_{k|K}. \quad (7)$$

2.4 Selecting the Number of Groups

To choose the optimal number of groups K , we use the following information criterion:

$$IC(K) = \log V_{N,T}^2(K) + K \cdot \rho, \quad (8)$$

where ρ is a penalty term depending on sample size and bandwidth, and

$$V_{N,T}^2(K) = \frac{1}{NT} \sum_{k=1}^K \sum_{i \in \hat{G}_{k|K}} \sum_{t=1}^T \left[Y_{it}^c - X_{it}^\top \hat{\gamma}_{k|K} \left(\frac{t}{T} \right) \right]^2 W \left(\frac{t}{T} \right). \quad (9)$$

The number of groups $\hat{K}$ is selected to minimize $IC(K)$ over $K = 1, \dots, \tilde{K}$, where $\tilde{K}$ is a user-specified maximum number of groups.

3 Replication of Chen

In this section we replicate the results from Chen starting with the data generating process to prove the consistnecy of our estimator, and following their empirical appplication on the gdp growth

3.1 DGP: Example 1

To establish a reliable foundation for our subsequent empirical analysis, we replicate the DGP simulation conducted by Chen, who in turn adopts the data generating process (DGP) proposed by Su, Shi, and Phillips (2018). Our objective here is not to extend Chen's methodological framework, but to verify its robustness and internal consistency through independent implementation. Specifically, we reproduce her simulation using Python rather than Matlab, thereby evaluating whether the estimator reliably identifies latent group structures and recovers time-varying coefficients under varying sample sizes and time periods.

Despite inevitable differences stemming from the stochastic generation of data and discrepancies in computational environments between Python and Matlab, our replication yields results that are qualitatively consistent with those reported by Chen. This consistency confirms the stability and robustness of the estimator across platforms and supports its suitability for applications involving grouped heterogeneity and dynamic effects. The replication also helps validate our subsequent empirical application by demonstrating that, under controlled conditions, the grouping structure and associated coefficient functions can be reliably recovered.

We replicate DGP 2 from Su et al. (2018), which is specified as:

$$Y_{it} = \beta_i + \alpha_{i,0} \left(1 - \frac{t}{T}\right) + \alpha_{i,1} \left(1 - \frac{t}{T}\right) X_{it} + \varepsilon_{it}, \quad i = 1, \dots, N; \quad t = 1, \dots, T,$$

where β_i and ε_{it} are independently drawn from the standard normal distribution $N(0, 1)$ and are mutually independent. The coefficient functions $\alpha_{i,0}(u)$ and $\alpha_{i,1}(u)$ vary across three latent groups G_1, G_2 , and G_3 , defined as follows:

$$\alpha_{i,0}(u) = \begin{cases} 3F(u; 0.5, 0.1), & i \in G_1, \\ 3[2u - 6u^2 + 4u^3 + F(u; 0.7, 0.05)], & i \in G_2, \\ 3[4u - 8u^2 + 4u^3 + F(u; 0.6, 0.05)], & i \in G_3, \end{cases}$$

$$\alpha_{i,1}(u) = \begin{cases} 3(2u - 4u^2 + 2u^3 + F(u; 0.6, 0.1)), & i \in G_1, \\ 3(u - 3u^2 + 2u^3 + F(u; 0.7, 0.04)), & i \in G_2, \\ 3(0.5u - 0.5u^2 + F(u; 0.4, 0.07)), & i \in G_3, \end{cases}$$

where the function $F(u; \mu, \sigma)$ denotes a logistic transformation:

$$F(u; \mu, \sigma) = \frac{1}{1 + \exp\left(-\frac{u-\mu}{\sigma}\right)}.$$

The cross-sectional sample is partitioned into three groups with relative sizes $N_1 = 0.3N$, $N_2 = 0.3N$, and $N_3 = 0.4N$. For identifiability, the intercept functions $\alpha_{i,0}(t/T)$ are demeaned such that:

$$\sum_{t=1}^T \alpha_{i,0}\left(\frac{t}{T}\right) = 0.$$

Simulations are conducted for sample sizes $N = 50$ and 100 , and time series lengths $T = 40$ and 80 . For each configuration, 200 replications are generated. Bandwidth selection for nonparametric estimation of $\alpha_i(\cdot)$ follows a leave-one-out cross-validation procedure, as discussed in Section 4.1. The Epanechnikov kernel is employed:

$$K(u) = \frac{3}{4}(1 - u^2)_+, \quad \text{where } (v)_+ = \max\{v, 0\}.$$

Clustering performance is evaluated using two standard metrics: purity and normalized mutual information (NMI), which compare the estimated cluster assignment $\hat{C} = \{\hat{G}_1, \dots, \hat{G}_K\}$ with the true partition $C^0 = \{G_1, \dots, G_{K^0}\}$. These are defined as:

$$\begin{aligned} \text{Purity}(\hat{C}, C^0) &= \frac{1}{N} \sum_{k=1}^{\hat{K}} \max_{1 \leq j \leq K^0} |\hat{G}_k \cap G_j|, \\ \text{NMI}(\hat{C}, C^0) &= \frac{I(\hat{C}; C^0)}{\frac{1}{2}[H(\hat{C}) + H(C^0)]}, \end{aligned}$$

where $I(\hat{C}; C^0)$ denotes mutual information and $H(\cdot)$ denotes entropy. Both measures are invariant to permutation of labels, with values closer to one indicating better clustering quality.

Estimation accuracy is evaluated using the root mean squared error (RMSE), calculated for three distinct estimators. The first is the *oracle estimator*, which assumes perfect knowledge of the true group membership. The second is the *pre-clustering estimator*, which estimates coefficients individually without incorporating any group membership information. Finally, the *post-clustering estimator* computes coefficient estimates based on the group structure inferred from the data, using group assignments selected according to the GAIC and GBIC criteria.

$$\text{RMSE}(\hat{\alpha}) = \sqrt{\frac{1}{N} \sum_{i=1}^N \frac{1}{T} \sum_{t=1}^T \left(\hat{\alpha}_i\left(\frac{t}{T}\right) - \alpha_i\left(\frac{t}{T}\right) \right)^2}.$$

Summary of Results

Table 1: Frequencies at which K_0 is estimated for Simulated Example 1

Sample Size		GBIC					GAIC				
		1	2	3 (true)	4	5	1	2	3 (true)	4	5
N = 50	T = 40	0	14	185	1	0	0	5	187	8	0
	T = 80	0	0	200	0	0	0	0	200	0	0
N = 100	T = 40	0	8	182	10	0	0	3	166	31	0
	T = 80	0	0	200	0	0	0	0	200	0	0

As seen in Table 1, the grouping estimator reliably identifies the correct number of clusters. GBIC successfully selects the true number of groups in over 92% of cases when $T = 40$ and in 100% of cases when $T = 80$, across both sample sizes. GAIC shows similar improvements with increasing T , though it performs slightly less consistently when $T = 40$. While numerical values differ slightly from those reported by Chen—due to differences in computational environments and randomization protocols—the results support the same qualitative conclusions. In some cases, the Python implementation even yields slightly improved identification rates under GAIC, reinforcing the robustness of the procedure.

Table 2: The average (standard deviation) NMI’s and purities for Simulated Example 1

Sample Size	GBIC NMI	GBIC Purity	GAIC NMI	GAIC Purity
N = 50, T = 40	0.8469 (0.0871)	0.9294 (0.0724)	0.8480 (0.0844)	0.9370 (0.0555)
N = 50, T = 80	0.9861 (0.0337)	0.9956 (0.0115)	0.9861 (0.0337)	0.9956 (0.0115)
N = 100, T = 40	0.8355 (0.0788)	0.9317 (0.0600)	0.8352 (0.0768)	0.9390 (0.0462)
N = 100, T = 80	0.9833 (0.0312)	0.9952 (0.0103)	0.9833 (0.0312)	0.9952 (0.0103)

Table 3: Avg. (SD) RMSE of $\beta_i(\cdot)$ estimates for Simulated Example 1

Sample Size	Oracle	Pre-clustering	Post (GBIC)	Post (GAIC)
N = 50, T = 40	0.2624 (0.0156)	0.4858 (0.0165)	0.3028 (0.0429)	0.2998 (0.0359)
N = 50, T = 80	0.1478 (0.0102)	0.3716 (0.0106)	0.1510 (0.0133)	0.1510 (0.0133)
N = 100, T = 40	0.2530 (0.0119)	0.4870 (0.0105)	0.2945 (0.0352)	0.2915 (0.0287)
N = 100, T = 80	0.1805 (0.0076)	0.3605 (0.0076)	0.1837 (0.0103)	0.1837 (0.0103)

As summarized in Tables 2 and 3, the clustering algorithm performs well in recovering latent group structure, with both NMI and purity increasing significantly as the time dimension grows. Moreover, the post-clustering estimator achieves RMSE values that are very close to those of the oracle estimator, and substantially lower than the pre-clustering estimator. These findings highlight the value of incorporating group-level information in improving estimation accuracy, further supporting the consistency and efficiency of the methodology under the simulated DGP.

3.2 Empirical application: GDP growth

In this section, we apply the kernel HAC method to an empirical study of economic growth, based on the work by Chen. The growth model under consideration is specified as:

$$GY_{i,t} = \alpha_i + \beta_{i,0}(t/T)GK_{i,t} + \beta_{i,2}(t/T)GPOP_{i,t} + \epsilon_{i,t}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (10)$$

where $GY_{i,t}$ denotes the annual GDP growth rate of country i in year t , $GK_{i,t}$ is the annual growth rate of capital formation, and $GPOP_{i,t}$ represents the annual growth rate of the population. All three variables are expressed in percentage terms.

Although annual labor input growth would be preferable to $GPOP_{i,t}$, it is substituted with population growth due to the limited availability of labor input data. The dataset, obtained from the World Bank's World Development Indicators (WDI), covers 61 countries over the period 1971–2016.

We estimate the functional coefficients $\beta_i(\cdot) = (\beta_{i,0}(\cdot), \beta_{i,2}(\cdot))^T$ using nonparametric kernel smoothing with the Epanechnikov kernel. The bandwidth is selected via leave-one-out cross-validation (LOOCV). The kernel HAC method is then used to classify the estimated $\beta_i(\cdot)$ functions into groups. The number of groups is selected using the information criteria introduced in Section 2.4. Both GAIC and GBIC identify four groups.

Figure 1 displays the estimated group-specific functional coefficients, and Figure 2 shows the corresponding grouped data. The group membership of countries is presented in Table 4.

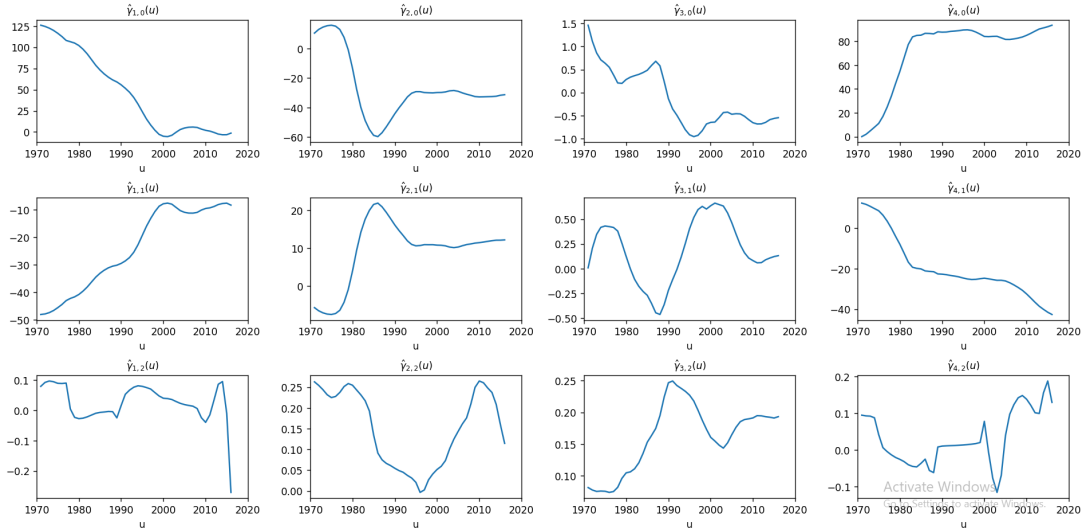


Figure 1: Post-clustering estimates of group-specific functional coefficients

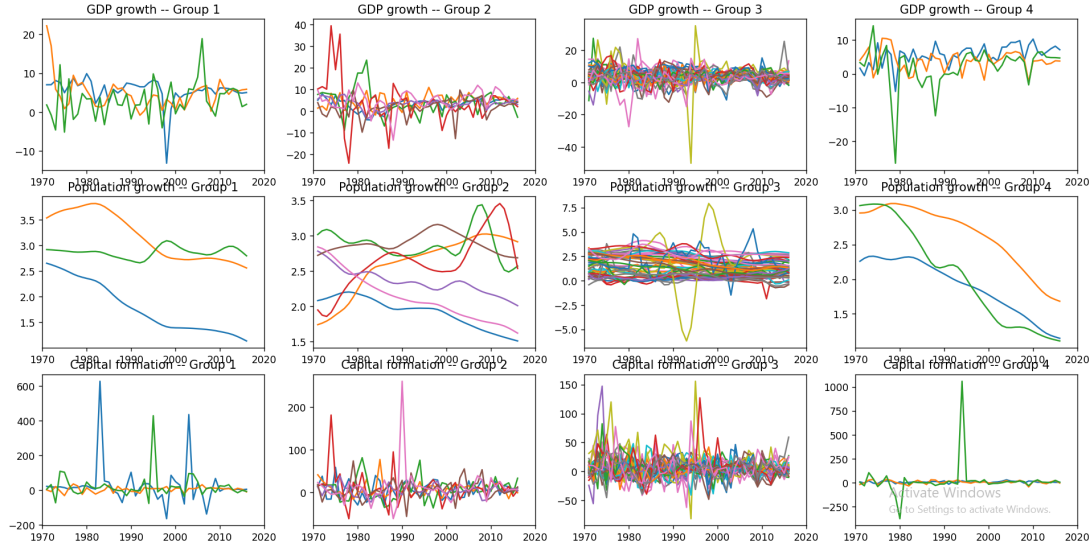


Figure 2: Data grouped by estimated clusters

Based on Table 4, the majority of countries are classified in Group 3. Groups 1 and 4 each comprise three members, while Group 2 includes seven countries. As shown in Figure 1, the estimated functional coefficients for Group 3 display less variation over time than those of the other groups, suggesting relatively stable economic structures. By contrast, Groups 1, 2, and 4 exhibit greater variability, which may indicate more substantial structural changes during the period.

The effect of capital formation growth on GDP also varies across groups. It is generally positive for Group 2, mixed for Group 3, and predominantly negative for Groups 1 and 4. Population growth shows a steadily positive impact in Group 2 since 2000, although this effect has declined since 2010. In Group 1, the impact of population growth on GDP appears to turn downward from 2015, becoming negative by 2017. This may reflect the context of these countries as developing economies with very high birth rates.

In contrast, in Group 4—which includes less-developed countries with high natality rates (e.g., Nicaragua)—the impact appears to remain positive and even to increase over time. This pattern suggests that underdeveloped countries beginning their growth trajectories may benefit from population growth.

Table 4: Group Membership of Countries

Group	Countries
Group 1	Indonesia, Kenya, Mauritania
Group 2	Bolivia, Burkina Faso, Congo, Rep., Gabon, Guatemala, Madagascar, Panama
Group 3	Argentina, Australia, France, Germany, Italy, Japan, Korea, Rep., Mexico, Netherlands, Spain, United Kingdom, United States, Algeria, Austria, Bangladesh, Belgium, Benin, Cameroon, Canada, Colombia, Denmark, Dominican Republic, Ecuador, Egypt, Arab Rep., Finland, Greece, Iran, Islamic Rep., Ireland, Lesotho, Luxembourg, Malaysia, Morocco, New Zealand, Norway, Pakistan, Peru, Philippines, Portugal, Rwanda, Senegal, Singapore, South Africa, Sri Lanka, Sudan, Sweden, Thailand, Togo, Uruguay
Group 4	India, Honduras, Nicaragua

4 Data

Having previously outlined the structural growth analysis, we now shift focus to our core contribution: a development-oriented model that goes beyond GDP growth to examine broader welfare dimensions. Specifically, we investigate two key outcomes—*GDP per capita* and the *poverty gap*—as dependent variables. This model incorporates a range of explanatory variables commonly associated with development outcomes, including *government spending*, *corruption*, and the *Human Development Index (HDI)*. Crucially, we also introduce *income inequality*, measured by the *Gini index*, as a moderating variable to assess how it shapes the effects of these factors.

The underlying dataset for this analysis is drawn from **Our World in Data**, covering indicators such as *GDP per capita*, *poverty gap*, *Gini index*, *corruption index*, *HDI*, and *government spending*. Although the initial dataset spans approximately **100 countries**, we restrict our final analysis to a balanced panel of **15 countries** from **1989 to 2022**, selected based on data availability, completeness, and consistency across all indicators. This restriction ensures robustness and comparability in the multivariate regressions and interaction analyses.

Our approach is twofold. First, we examine how government spending, corruption, and HDI affect *GDP per capita*, providing insight into average development outcomes. Second, we assess how the same set of regressors influence the *poverty gap*, capturing the depth of deprivation among the most vulnerable segments of the population.

A key innovation in our model is the inclusion of interaction terms between inequality and each explanatory variable. This allows us to investigate whether inequality conditions or amplifies the effects of development inputs. We hypothesize that while inequality may exert a limited influence on the relationship between the regressors and *GDP per capita*—which reflects average national performance—it will play a far more significant role in shaping the determinants of the *poverty gap*, which is sensitive to distributional dynamics.

In essence, by incorporating inequality as a moderating factor, we aim to reveal differentiated effects that are often concealed by aggregate measures. We expect that these interaction effects will show how factors like government spending and corruption can have heterogeneous

impacts depending on the level of inequality—insights that are especially important for designing inclusive development policies.

4.1 Missing Values

A common approach to handling missing data is *linear interpolation*, which estimates missing values by connecting known data points with straight lines. However, due to the high proportion of missing observations in our dataset, linear interpolation often results in inaccurate imputations. In particular, some values expected to be positive based on economic reasoning are imputed as negative, thereby undermining the dataset’s validity.

Following this failed attempt, we also considered *Kalman filtering*, introduced by Kalman (1960), a widely used state-space method for estimating unobserved components in time series data. Despite our best efforts, we ultimately opted out of Kalman *smoothing*, as it tends to dampen short-run fluctuations that are crucial for our analysis. Our aim is to preserve the natural variation inherent in macroeconomic series.

To overcome these challenges, we implement a more advanced technique known as *cubic Hermite spline interpolation*, following the methodology developed by Fritsch & Carlson (1980). This method preserves the shape and monotonicity of the original data while providing a smooth interpolation curve. Its key advantage lies in balancing the trade-off between maintaining monotonicity and accurately approximating the missing values.

The interpolation procedure involves two stages:

Step 1: Initialization of Derivatives

First, initial derivatives d_i are assigned at each data point (x_i, y_i) to ensure consistency with the local slopes. For $i = 1, 2, \dots, n$, the signs of d_i and d_{i+1} are matched to the sign of the secant slope over the interval $I_i = [x_i, x_{i+1}]$:

$$\text{sgn}(d_i) = \text{sgn}(d_{i+1}) = \text{sgn}(\Delta_i),$$

where

$$\Delta_i = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}.$$

If $\Delta_i = 0$, then $d_i = d_{i+1} = 0$, ensuring that the interpolant remains constant over the interval.

Step 2: Monotonicity Adjustment

Next, the initial derivatives are adjusted as necessary to guarantee that the cubic interpolant remains monotonic on each subinterval. For any interval I_i where the pair $(\alpha_i, \beta_i) \notin \mathcal{M}$, the derivatives d_i and d_{i+1} are modified to new values d_i^* and d_{i+1}^* , ensuring that

$$\alpha_i^* = \frac{d_i^*}{\Delta_i}, \quad \beta_i^* = \frac{d_{i+1}^*}{\Delta_i} \in \mathcal{M},$$

where $\mathcal{M}$ denotes the set of admissible values preserving monotonicity.

It is important to note that modifying d_i affects not only the interval I_i but also its adjacent interval I_{i-1} , since

$$\beta_{i-1}\Delta_{i-1} = d_i = \alpha_i\Delta_i.$$

Therefore, care must be taken to maintain monotonicity consistently across all intervals. One strategy involves defining a subset $\mathcal{S} \subset \mathcal{M}$ with properties that ensure the stability of adjusted derivatives throughout the data.

By applying monotone piecewise cubic interpolation, we obtain a smooth and economically meaningful imputation of missing values, preserving important features such as monotonic trends and short-term variability. This method aligns with best practices in panel data imputation and enhances the reliability of subsequent econometric analyses.

4.2 Statistics

Although our dataset spans only 35 years and includes 15 countries, it offers valuable insights into the evolving relationships between key economic indicators. To investigate these dynamics, we begin by analyzing developed economies, visualizing the behavior of selected regressors in two benchmark years—1989 and 2022. Specifically, we plot GDP per capita and the poverty gap against various regressors to discern changes over time. We then apply the same methodology to developing countries, enabling us to compare the relationships across the development spectrum.

In developing countries, GDP per capita in the early stages of economic development tends to exhibit a positive correlation with income inequality, as measured by the Gini coefficient. In contrast, this relationship appears to plateau in developed nations. These patterns align with the early phase of the Kuznets Curve, which posits that inequality initially rises with economic growth before stabilizing or declining. However, contrary to the curve’s original hypothesis, our findings suggest that inequality continues to rise with GDP per capita in both developing and developed countries. This persistent trend challenges the notion that economic growth inherently leads to more equitable outcomes and underscores the importance of understanding how inequality modulates the effects of development.

In high-income economies, increases in GDP per capita may disproportionately benefit upper-income groups, resulting in stagnant or even increasing inequality. This pattern supports our central thesis: inequality is not merely a byproduct of development but may actively shape the nature and distribution of its benefits.

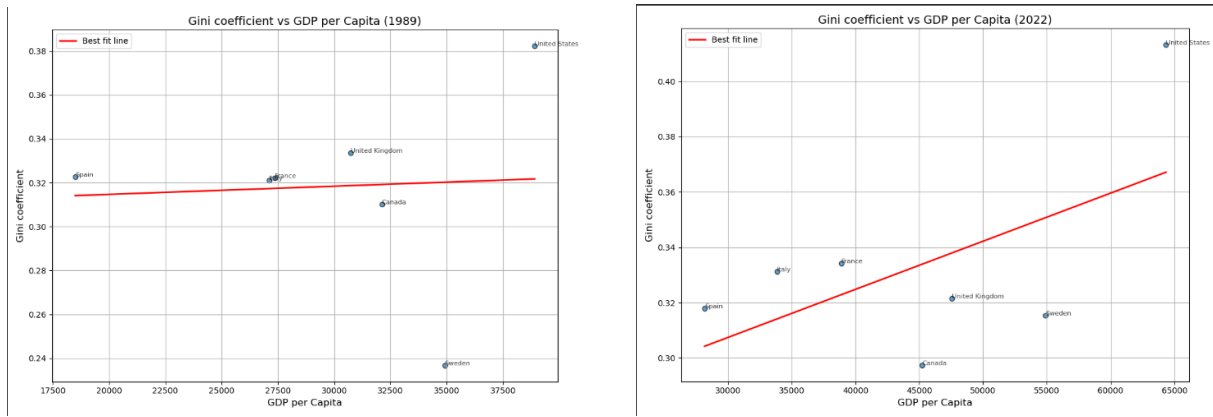


Figure 3: Gini Coefficient against GDP per capita in developed countries between 1989 (left) and 2022 (right).

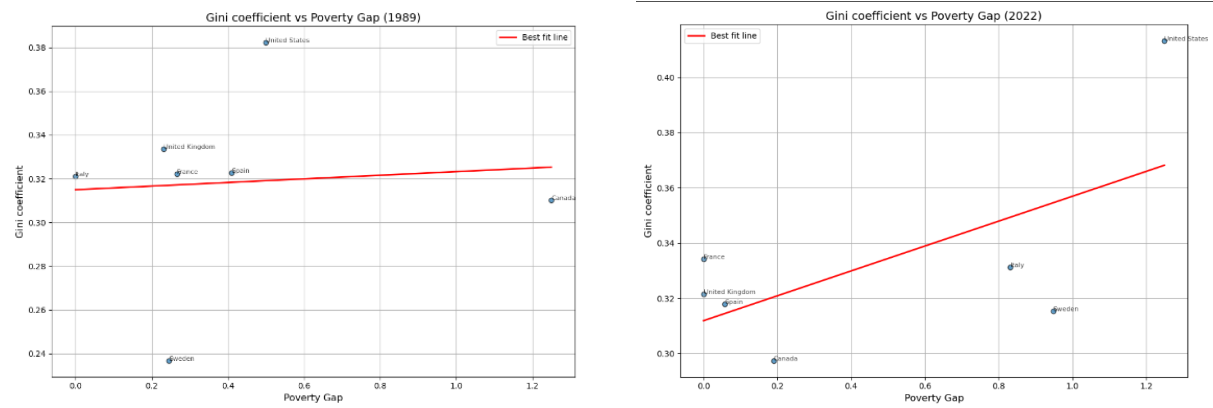


Figure 4: Gini Coefficient against Poverty gap in developed countries between 1989 (left) and 2022 (right).

In 1989, the relationship between the poverty gap and inequality appeared relatively weak in both developed and developing countries, suggesting that poverty, rather than inequality, was the more salient challenge at the time. However, in 2022, this relationship becomes markedly more pronounced across both groups. The strong positive correlation now observed between the Gini coefficient and the poverty gap implies that inequality may be a key driver of poverty itself.

This observation raises important concerns. As inequality increases, access to essential resources such as quality education, healthcare, and social mobility becomes more limited. The result is a self-perpetuating poverty trap: individuals may continue to invest effort and labor without seeing commensurate improvements in their economic circumstances. The erosion of belief in upward mobility further exacerbates this cycle, contributing to persistent intergenerational inequality.

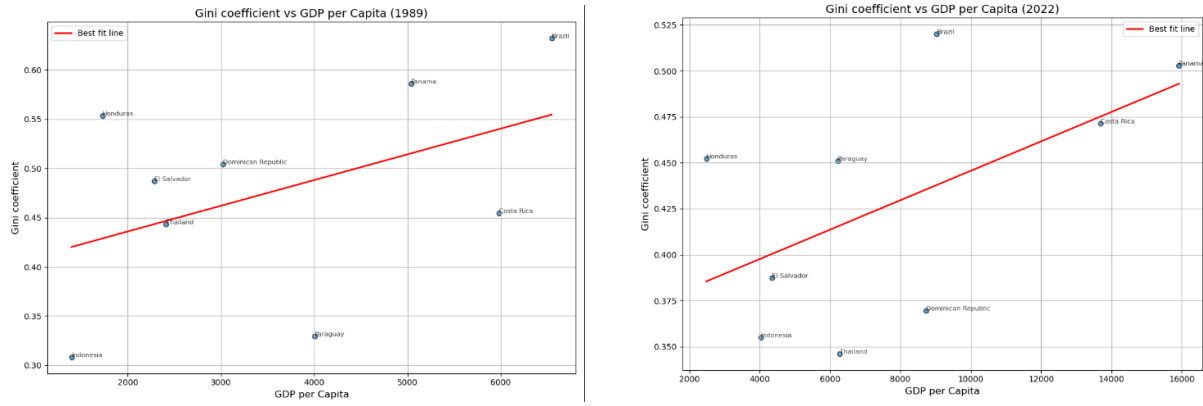


Figure 5: Gini Coefficient against GDP per capita in developing countries between 1989 (left) and 2022 (right).

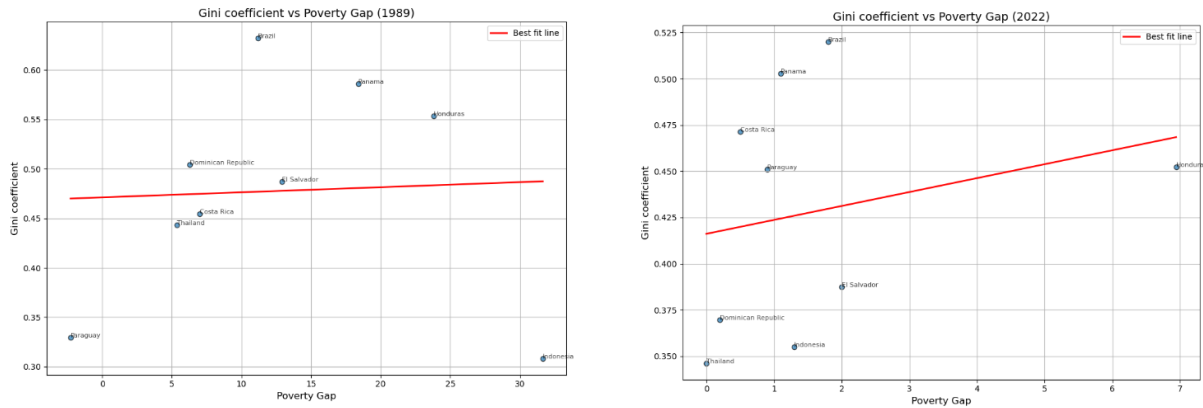


Figure 6: Gini Coefficient against Poverty gap in developing countries between 1989 (left) and 2022 (right).

These findings emphasize a central insight of this study: to fully grasp the dynamics of economic development, one must look beyond average indicators such as GDP per capita. While economic growth is a necessary condition—particularly for low-income countries—it is not a sufficient one (Stiglitz et al., 2009). The statistics suggest that inequality plays a mediating role in determining how other factors, such as human development, governance quality, and fiscal policy, translate into tangible improvements in welfare (Ravallion, 2012; Ostry et al., 2014).

In particular, the relationship between inequality and the Human Development Index (HDI) deserves attention. Although GDP per capita and HDI are positively correlated, the benefits of enhanced education, health, and living standards may be unevenly distributed in highly unequal societies. In such contexts, gains in HDI may fail to reach the most vulnerable populations, thereby limiting the developmental impact of aggregate improvements (UNDP, 2019; Deaton, 2013).

Similarly, inequality appears to condition the effectiveness of government spending. In more equitable societies, public expenditures on health, education, and infrastructure are more likely to foster inclusive growth (Gupta et al., 2002). However, in contexts of high inequality, these investments may disproportionately benefit privileged groups, reinforcing existing disparities rather than reducing them (Alesina & Rodrik, 1994; Milanovic, 2016).

Finally, the interaction between inequality and corruption must also be considered. In unequal societies, governance institutions may be more susceptible to capture by elites, weakening the rule of law and distorting public resource allocation (Acemoglu & Robinson, 2006; Chong & Gradstein, 2007). This dynamic not only hampers development but also erodes public trust and institutional legitimacy, further entrenching cycles of poverty and exclusion.

Taken together, these patterns suggest that inequality is not merely a parallel outcome of economic processes but a central variable that shapes how other determinants of development—such as HDI, corruption, and public spending—manifest in practice. Addressing inequality, therefore, is essential for ensuring that growth and institutional improvements translate into broadly shared progress.

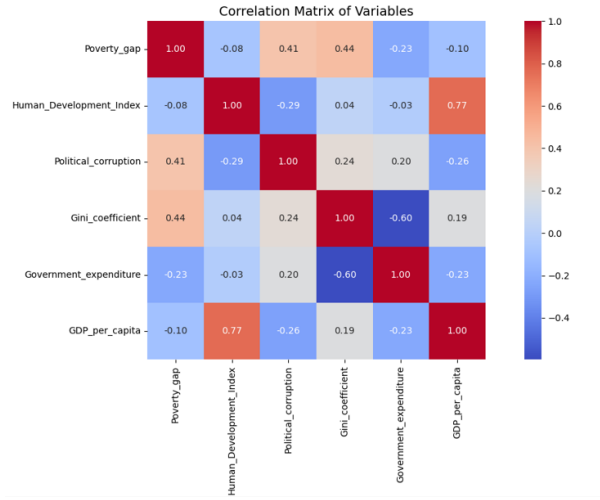


Figure 7: Pearson Correlation matrix

Assessing the Pearson correlation matrix, which pools all variables across years regardless of entities, we observe several notable patterns. As expected, there is a positive overall correlation between the Gini coefficient and GDP per capita. This suggests a general trend where increases in GDP per capita are associated with higher levels of inequality. Similarly, the relationship between inequality and the poverty gap is also positive, indicating that higher poverty levels tend to coincide with greater inequality. Another significant correlation is observed between GDP per capita and the Human Development Index (HDI), which is relatively strong. This implies that components of the HDI—such as education and life expectancy—are closely linked to GDP per capita. However, since none of the correlation coefficients exceed 0.8, multicollinearity is not a concern, and we can include all regressors in our panel data model without issue.

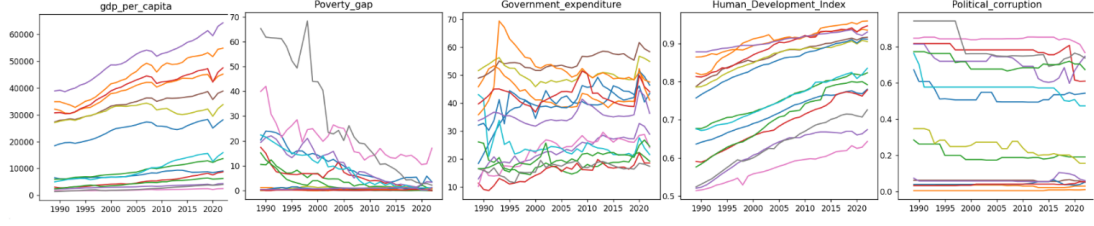


Figure 8: Evolution over time of the regressors and dependent variables without controlling for inequality. The plots illustrate general trends in GDP per capita, poverty gap, Human Development Index, government expenditure, and corruption index, highlighting overall progress despite persistent challenges.

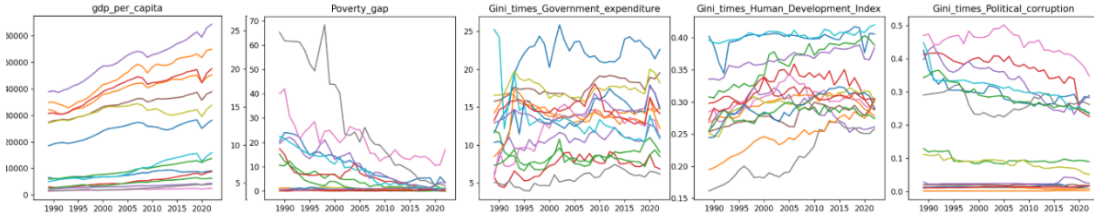


Figure 9: Evolution over time of the interaction terms (Gini coefficient $\times$ regressors) and dependent variables, capturing the impact of inequality. These plots show increased fluctuation and loss of monotonic trends, suggesting that inequality modifies the relationship between key drivers and development outcomes.

Moving from the pooled interaction to the evolution of variables over time—both without and with accounting for inequality—we observe in Figure 8 several important trends. GDP per capita has generally increased over the period, while the poverty gap has narrowed, suggesting meaningful progress in reducing extreme poverty and raising overall income levels. The Human Development Index shows a generally monotonic upward trend, reflecting broad improvements in health, education, and living standards. Government expenditure, while more volatile year to year, appears to trend upward as well, possibly indicating greater investment in public goods and services over time. In contrast, the corruption index remains relatively flat, with only minor fluctuations, suggesting that while economic and social indicators have improved, institutional quality—at least as measured by corruption—has seen limited progress and remains a persistent challenge.

However, when we account for inequality by interacting the regressors with the Gini coefficient, the evolution of these variables over time becomes more erratic and less monotonic. This greater fluctuation indicates that inequality shapes how these improvements are distributed: increases in GDP per capita or government spending may not translate evenly across all groups. For instance, gains in human development or government expenditure might disproportionately benefit better-off segments of the population, leaving poorer groups behind. Similarly, the link between corruption and development outcomes may be more complex when inequality is taken into account, with corruption potentially reinforcing or exacerbating unequal access to the benefits of growth.

These findings highlight the importance of considering inequality explicitly in our analysis. In the next section, we apply our proposed methodology, based on Chen’s approach, to estimate

the functional coefficients while accounting for heterogeneity across countries. This will allow us to identify groups of countries that follow similar development paths and better understand the role of inequality in shaping these trajectories.

5 Results

To assess the dynamic relationship between development factors and socioeconomic outcomes, we estimate four model specifications. Models 1 and 2 focus on GDP per capita as the outcome, while Models 3 and 4 examine the poverty gap. Within each pair, the first specification (Models 1 and 3) includes only the direct effects of the explanatory variables, allowing their impact to vary smoothly over time. In contrast, the second specification (Models 2 and 4) incorporates interactions between each regressor and the Gini coefficient, enabling us to capture how the influence of development factors may depend on a country’s level of income inequality. This comparison allows us to distinguish between general trends and inequality-contingent effects in shaping economic growth and poverty outcomes over time.

Let $GDP_{i,t}$ denote GDP per capita and $PovGap_{i,t}$ the poverty gap for country i at time t . The regressors are human development index ($X_1 = HDI_{i,t}$), corruption perception ($X_2 = Corruption_{i,t}$), and government spending ($X_3 = GovSpend_{i,t}$). Income inequality is measured by the Gini coefficient ($Gini_{i,t}$). The coefficients $\beta_j(\cdot)$ and $\delta_j(\cdot)$ vary smoothly over normalized time $\frac{t}{T}$, where T is the total number of periods. Fixed effects for countries are captured by α_i .

$$\textbf{Model 1: } GDP_{i,t} = \alpha_i + \sum_{j=1}^3 \beta_j\left(\frac{t}{T}\right) X_{j,i,t} + \epsilon_{i,t} \quad (11)$$

$$\textbf{Model 2: } GDP_{i,t} = \alpha_i + \sum_{j=1}^3 \delta_j\left(\frac{t}{T}\right) (X_{j,i,t} \cdot Gini_{i,t}) + \epsilon_{i,t} \quad (12)$$

$$\textbf{Model 3: } PovGap_{i,t} = \alpha_i + \sum_{j=1}^3 \beta_j\left(\frac{t}{T}\right) X_{j,i,t} + \epsilon_{i,t} \quad (13)$$

$$\textbf{Model 4: } PovGap_{i,t} = \alpha_i + \sum_{j=1}^3 \delta_j\left(\frac{t}{T}\right) (X_{j,i,t} \cdot Gini_{i,t}) + \epsilon_{i,t} \quad (14)$$

This framework captures how the influence of key development indicators evolves over time and how these effects interact with inequality through the Gini coefficient.

Analysis of Group Membership Before and After Inequality Adjustment

When analyzing the group membership of countries under the baseline specification—where the effects of development variables are modeled without interaction with the Gini coefficient—we observe several meaningful patterns. Despite limiting the analysis to countries with relatively complete data, the majority of countries in both the GDP per capita and poverty gap specifications are clustered into a single dominant group. This suggests that countries with similar levels of development or economic structure may share comparable growth or poverty dynamics, although substantial heterogeneity remains.

Looking more closely, we notice a divergence in the composition of this dominant group de-

pending on the dependent variable. In the GDP per capita model, the group is largely composed of low- to middle-income countries such as Brazil, Indonesia, Honduras, and El Salvador. These countries appear to be following similar growth trajectories, possibly due to their comparable positions in the development process—characterized by structural transformation, industrialization, and improving institutional frameworks. The impact of development variables such as human development (HDI), government spending, and corruption perception on GDP per capita seems broadly consistent across these countries, leading to shared clustering patterns.

In contrast, the poverty gap model, without inequality adjustment, places many high-income countries such as Canada, France, the United Kingdom, and the United States into the same group. This outcome is somewhat surprising, as these countries differ in terms of welfare state models and fiscal policy. However, it may suggest that in the absence of inequality considerations, high-income countries share similar structural characteristics that affect poverty reduction uniformly. Conversely, developing countries are more dispersed across clusters, indicating greater heterogeneity in how development indicators translate into poverty outcomes. This dispersion may reflect fundamental structural challenges—such as lack of access to education, weak redistributive systems, high levels of corruption, or limited fiscal capacity—that lead to diverse paths of poverty reduction.

Introducing inequality via interaction terms between the Gini coefficient and each regressor substantially alters this clustering landscape—but asymmetrically across the two outcomes. In the GDP per capita model, group membership remains largely stable. Countries that were clustered together without inequality adjustment tend to remain in the same groups after the Gini interactions are included. This stability reflects the nature of GDP per capita as an aggregate measure of income, which is relatively insensitive to its distribution. While inequality may have long-run macroeconomic implications, its short-run interaction with development factors has a muted effect on average income levels. As a result, countries with similar economic fundamentals remain clustered together, even when inequality is accounted for.

By contrast, the poverty gap model displays significant changes in group membership after inequality adjustment. Countries that were previously grouped together based on similar development profiles are now reassigned to different clusters. This disruption highlights the poverty gap’s sensitivity to income distribution: unlike GDP per capita, it captures the depth and severity of poverty. Introducing the Gini interaction allows the model to account for how development indicators affect the poor differently depending on the degree of inequality. For instance, high government spending in an unequal country may not reach the poorest households effectively, resulting in higher poverty persistence despite favorable development indicators. Thus, two countries with similar HDI or spending levels may diverge significantly in their poverty outcomes when inequality is taken into account.

This divergence is especially pronounced among developing countries, where structural differences—such as limited access to education, inefficient redistribution, or corruption—can significantly moderate the impact of development variables. Consequently, the inclusion of inequality in the poverty model reveals greater heterogeneity, reshaping the cluster composition and offering a more nuanced view of development outcomes.

In summary, while the GDP per capita model exhibits relatively stable groupings before and

after accounting for inequality, the poverty gap model undergoes substantial reorganization. This finding underscores a key insight: inequality plays a limited role in shaping aggregate income dynamics, but has a profound effect on the effectiveness of development policies aimed at poverty reduction. Accordingly, incorporating distributional factors into models of poverty is essential for understanding the true variation in how development is experienced across countries.

Conclusion

This analysis has applied clustering techniques of the time-varying coefficients estimating using K-smooth functions to group countries based on economic indicators across four models—GDP per capita and poverty gap, each with and without inequality adjustment. Despite a significantly reduced sample of 15 countries due to missing data, several meaningful insights have emerged.

First, we have observed that in GDP per capita models, developing countries often clustered together, suggesting commonalities in their growth patterns. However, when inequality was taken into account, some high-income countries such as France and Spain have shifted into lower-performing groups, indicating that inequality can significantly alter perceived economic outcomes.

In contrast, the poverty gap models—especially when adjusted for inequality—revealed a clearer pattern: high-income countries tended to cluster together, suggesting similar dynamics in how poverty is addressed or experienced across these nations. Meanwhile, developing countries were widely dispersed across different clusters, reflecting diverse challenges such as corruption, limited social spending, and lack of access to education or healthcare.

These findings imply that while economic growth (as measured by GDP per capita) may follow somewhat consistent paths across similar income levels, poverty outcomes are more sensitive to both structural and institutional differences, particularly in developing nations.

A key limitation of this study is the restricted sample size, driven by data availability. As such, these groupings may not be fully generalizable. Future research should aim to include a broader set of countries and explore additional variables to capture more dimensions of development and inequality. Furthermore, maybe introducing a sieve estimation may actually lead to different clusters and maybe different interpretations, suggesting that this could be an area of further research.

Ultimately, the results highlight the importance of tailored, context-specific approaches to poverty reduction. While developed countries may benefit from similar policy tools, developing countries require nuanced strategies that address their unique socio-economic challenges.

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A Programming code

The coding was done in Python and more about the implementation can be found in the redame file.

B Tables

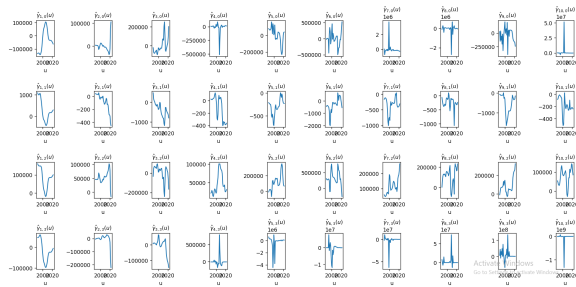


Figure 10: Post-clustering estimates of group-specific functional coefficients

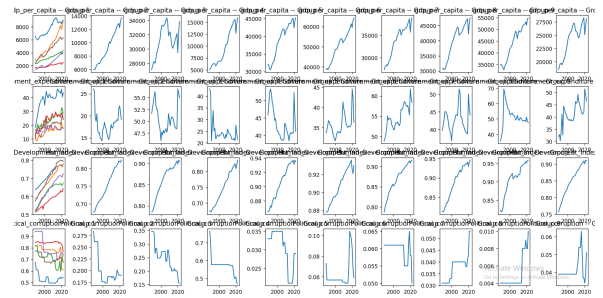


Figure 11: A plot of data by estimated groups

Model specification with GDP per capita as dependent variable and no control for inequality

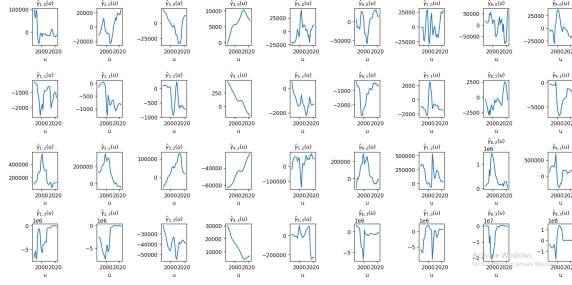


Figure 12: Post-clustering estimates of group-specific functional coefficients

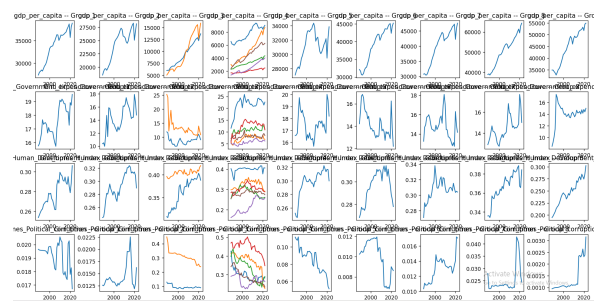


Figure 13: A plot of data by estimated groups

Model specification with GDP per capita as dependent variable and control for poverty (regressors are interacted with Gini coefficient)

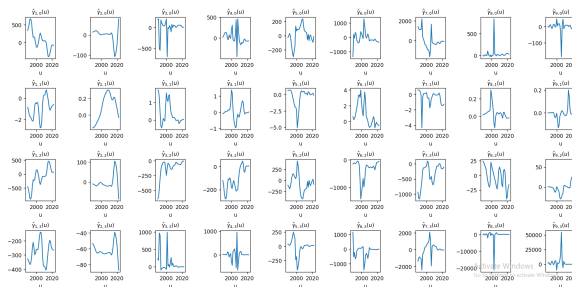


Figure 14: Post-clustering estimates of group-specific functional coefficients

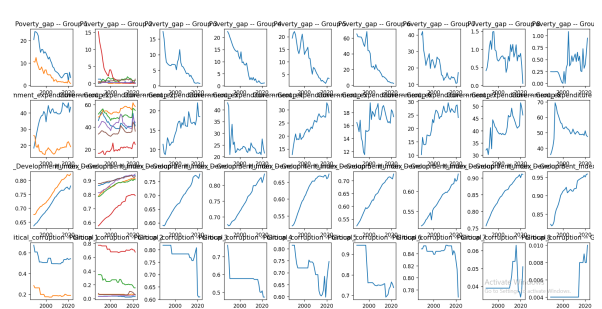


Figure 15: A plot of data by estimated groups

Model specification with poverty rate as dependent variable and no control for inequality

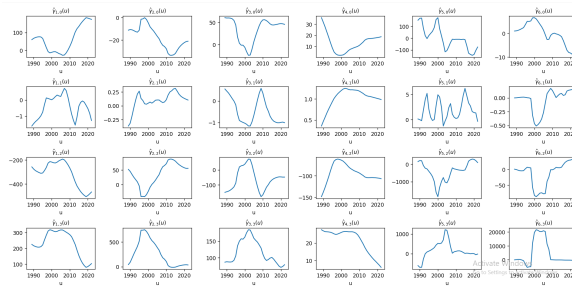


Figure 16: Post-clustering estimates of group-specific functional coefficients

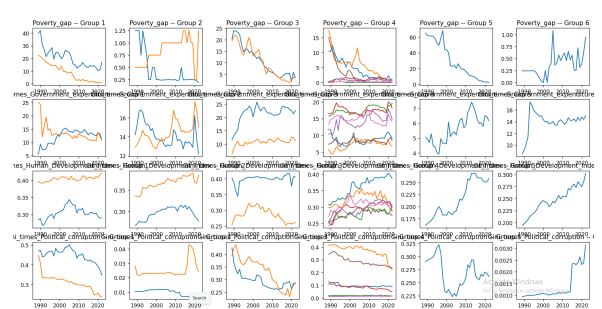


Figure 17: A plot of data by estimated groups

Model specification with poverty gap as dependent variable and control for inequality (regressors are interacted with Gini coefficient)

Group	Countries
1	Brazil, Dominican Republic, El Salvador, Honduras, Indonesia, Thailand
2	Costa Rica
3	Italy
4	Panama
5	Canada
6	United States
7	France
8	United Kingdom
9	Sweden
10	Spain

Table 5: Model 1: GDP per capita (no inequality adjustment)

Group	Countries
1	France
2	Spain
3	Costa Rica, Panama
4	Brazil, Dominican Republic, El Salvador, Honduras, Indonesia, Thailand
5	Italy
6	Canada
7	United Kingdom
8	United States
9	Sweden

Table 6: Model 2: GDP per capita (with inequality adjustment)

Group	Countries
1	Brazil, Costa Rica
2	Canada, France, Italy, Thailand, United Kingdom, United States
3	Dominican Republic
4	Panama
5	El Salvador
6	Indonesia
7	Honduras
8	Spain
9	Sweden

Table 7: Model 3: Poverty gap (no inequality adjustment)

Group	Countries
1	Honduras, Panama
2	Canada, United States
3	Brazil, El Salvador
4	Costa Rica, Dominican Republic, France, Italy, Spain, Thailand, United Kingdom
5	Indonesia
6	Sweden

Table 8: Model 4: Poverty gap (with inequality adjustment)